## DEFINITIONS

J = joule
kWh = kilowatt-hour
hp = horsepower
C = coulomb
s = second
A = ampere ("amp")
V = voltage ("volt")
W = watt

## CONSTANTS

Electron Charge = $-1.602\times10^{-19}$ C
Proton Charge = $+1.602\times10^{-19}$ C

## BASIC CONVERSIONS

1 kWh = $3.6 \times 10^6$ J
1 hp = 745.7 W
1 A = 1 C/s
1 V = 1 J/C
1 W = 1 J/s = 1 V-A

## BASIC UNITS

| | | | | | |
|---|---|---|---|---|---|
| $10^{-12}$ | pico | p | $10^{0}$ | - | - |
| $10^{-9}$ | nano | n | $10^{3}$ | kilo | k |
| $10^{-6}$ | micro | μ | $10^{6}$ | mega | M |
| $10^{-3}$ | milli | m | $10^{9}$ | giga | G |

## OHM'S LAW: $V = IR,\ I = {}^{V}/_{R},\ R = {}^{V}/_{I}$

## BASIC POWER (DC)

**Sources**: $P = VI$, sources can deliver or absorb power

**Resistors**: $P = VI = I^2R = {}^{V^2}/_{R}$, only absorb power

**Power Efficiency**: $\eta = \frac{P_{OUT}}{P_{IN}} x100\%$

## RESISTORS IN SERIES

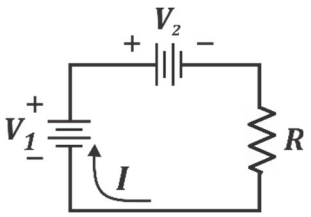
$$R_{equivalent} = R_1 + R_2 + R_3 + \cdots + R_N$$

## RESISTORS IN PARALLEL

$$\frac{1}{R_{equivalent}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_N}$$

## KIRCHHOFF'S VOLTAGE LAW

$$\sum_{drops}^{loop} \boldsymbol{V_i} = \mathbf{0}$$




**EXAMPLE:**

$$-V_1 + V_2 + IR = 0$$

**OR**

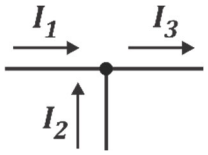
$$\sum_{rises}^{loop} \boldsymbol{V_i} = \mathbf{0}$$

**EXAMPLE:**

$$+V_1 - V_2 - IR = 0$$

## KIRCHHOFF'S CURRENT LAW

$$\sum_{in}^{node} \boldsymbol{I_i} = \mathbf{0}$$




**EXAMPLE:**

$$I_1 + I_2 - I_3 = 0$$

**OR**

$$\sum_{in}^{node} \boldsymbol{I_i} = \sum_{out}^{node} \boldsymbol{I_j}$$

**EXAMPLE:**

$$I_1 + I_2 = I_3$$

## NODAL ANALYSIS

1. Select a reference node (or ground node)
2. Label nodes with voltages, i.e. $V_1, V_2, \ldots, V_{N-1}$
3. Write KCL equations for (N-1) nodes*
4. Solve for N-1 voltages

* May need SuperNode if embedded voltage source connects two nodes

## MESH ANALYSIS

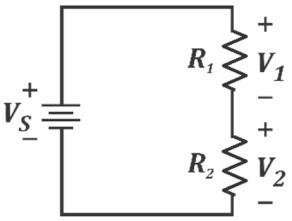
1. Check that circuit is planar (always in ECE2004)
2. Label mesh currents i.e. $I_1, I_2, \ldots, I_M$ *(CW or CCW, your discretion)*
3. Write KVL equations for M meshes*
4. Solve for M mesh currents

* May need SuperMesh if embedded current source exists between two meshes

## VOLTAGE DIVIDER EQUATION

$$V_1 = V_S \frac{R_1}{(R_1 + R_2)}$$

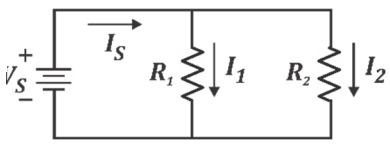
$$V_2 = V_S \frac{R_2}{(R_1 + R_2)}$$




## CURRENT DIVIDER EQUATION

$$I_1 = I_S \frac{R_2}{(R_1 + R_2)}$$

$$I_2 = I_S \frac{R_1}{(R_1 + R_2)}$$



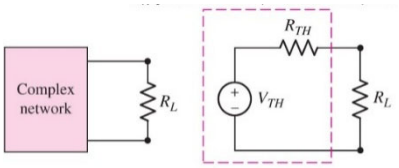
## SUPERPOSITION

1. Leave one *independent* source ON and set all other *independent* sources to zero* (*V = 0 is a short, I = 0 is open)*
2. Find the circuit response from this source
3. Repeat for all other independent sources
4. Add the results to find the total response

* Note: dependent sources are always left ON

## THEVENIN EQUIVALENT

1. Disconnect the load
2. Find the open circuit voltage $V_{oc}$ across open terminals
3. Find the equivalent resistance $R_{eq}$ into the network with all *independent* sources set to zero
4. $V_{TH} = V_{oc}$ and $R_{TH} = R_{eq}$

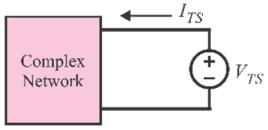



**Notes for finding $R_{TH}$ with Dependent Sources:** dependent sources are always left ON. With dependent sources, find $R_{TH}$ using the following one of the following techniques.

- Find $V_{TH}$ (Thevenin equivalent)
- Find $I_N$ (Norton equivalent)
- Solve $R_{TH}=V_{TH}/I_N$

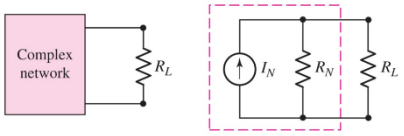
**OR (Test Source Approach)**

- Turn off all *independent* sources
- Attach "test source", $V_{TS}$, at terminals
- Find "test source current", $I_{TS}$
- Solve $R_{TH} = V_{TS}/I_{TS}$



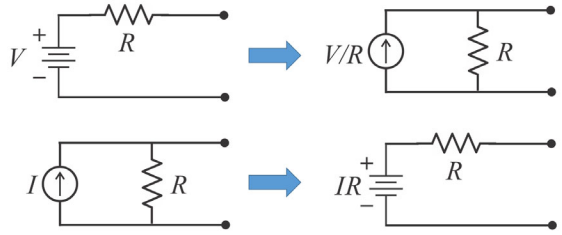

## NORTON EQUIVALENT

1. Replace the load terminals with a short
2. Find the current $I_{sc}$ through this short
3. Find the equivalent resistance $R_{eq}$ into the network with all *independent* sources set to zero
4. $I_N = I_{sc}$ and $R_N = R_{eq}$




**Notes for finding $R_{TH}$ with Dependent Sources**: dependent sources are always left ON. With dependent sources, find $R_{TH}$ using the same process as described above under Thevenin Equivalent.

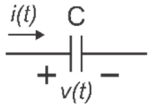
## SOURCE TRANSFORMATION PAIRS




Note: Arrow head points in the direction of "+"

## CAPACITORS

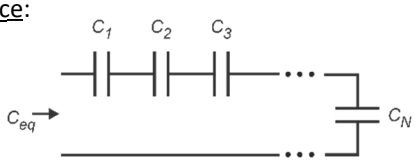
- Charge on capacitor plate (Coulombs) $= Q = CV$
- Capacitance based on physical parameters $= C = {}^{\varepsilon A}/_{d}$ with $\varepsilon = \varepsilon_0\varepsilon_r$ and $\varepsilon_0 = 8.854 \times 10^{-12}({}^{F}/_{m})$
- $i(t) = C\,{}^{dv(t)}/_{dt}$




- Voltage across a capacitor cannot change instantaneously
- Energy Storage on Capacitor $w(t) = \frac{1}{2}Cv^2(t)$

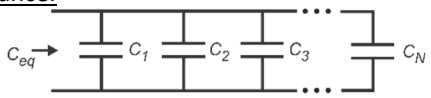
Equivalent Series Capacitance:

$$C_{eq} = \frac{1}{{}^{1}/_{C_1} + {}^{1}/_{C_2} \cdots {}^{1}/_{C_N}}$$



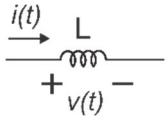

Equivalent Parallel Capacitance:

$$C_{eq} = C_1 + C_2 + \cdots + C_N$$




## INDUCTORS

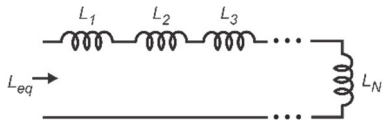
- Charge on capacitor plate (Coulombs) $= Q = CV$
- Inductance based on physical parameters with $\mu = \mu_0\mu_r$ and $\mu_0 = 4\pi \times 10^{-}({}^{H}/_{m})$
- $v(t) = L\,{}^{di(t)}/_{dt}$




- Current through an inductor cannot change instantaneously
- Energy Storage on Inductor $w(t) = \frac{1}{2}Li^2(t)$

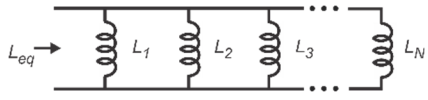
Equivalent Series Inductance:

$$L_{eq} = L_1 + L_2 + \cdots + L_N$$




Equivalent Parallel Inductance:

$$L_{eq} = \frac{1}{{}^{1}/_{L_1} + {}^{1}/_{L_2} \cdots {}^{1}/_{L_N}}$$



## TRANSIENT ANALYSIS OF RC NETWORKS

1) $t = 0^-$ (switch in starting position), Perform DC analysis with C replaced by "open" ($I_C(0^-) = 0$). Find $V_C(0^-)$ and other required voltages and currents.
2) $t = \infty$ (switch in final position), Perform DC analysis with C replaced by "open" ($I_C(\infty) = 0$). Find $V_C(\infty)$ and other required voltages and currents.
3) $t = 0^+$ (switch in final position), Perform DC analysis with C replaced by voltage source with $V_C(0^+) = V_C(0^-)$. Find $I_C(0^+)$ and other required voltages and currents.
4) Find $R_{EQ}$ across C (switch in final position). Calculate time constant, $\tau = R_{EQ}C$
5) Complete these equations for required voltages and currents

$v(t) = V(\infty) + [V(0^+) - V(\infty)]e^{-t/\tau}$

$i(t) = I(\infty) + [I(0^+) - I(\infty)]e^{-t/\tau}$

## TRANSIENT ANALYSIS OF RL NETWORKS

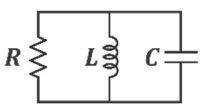
1) $t = 0^-$ (switch in starting position), Perform DC analysis with L replaced by "short" ($V_L(0^-) = 0$). Find $I_L(0^-)$ and other required voltages and currents.
2) $t = \infty$ (switch in final position), Perform DC analysis with L replaced by "short" ($V_L(\infty) = 0$). Find $I_L(\infty)$ and other required voltages and currents.
3) $t = 0^+$ (switch in final position), Perform DC analysis with L replaced by current source with $I_L(0^+) = I_L(0^-)$. Find $V_L(0^+)$ and other required voltages and currents.
4) Find $R_{EQ}$ across L (switch in final position). Calculate time constant, $\tau = {L}/{R_{EQ}}$
5) Complete these equations for required voltages and currents

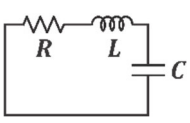
$v(t) = V(\infty) + [V(0^+) - V(\infty)]e^{-t/\tau}$

$i(t) = I(\infty) + [I(0^+) - I(\infty)]e^{-t/\tau}$

## TRANSIENT ANALYSIS OF RLC NETWORKS

1) $t = 0^-$ (switch in starting position), Perform DC analysis with C replaced by "open" ($I_C(0^-) = 0$) and L replaced by "short" ($V_L(0^-) = 0$). Find $V_C(0^-)$ and $I_L(0^-)$ and other required voltages and currents.
2) $t = \infty$ (switch in final position), Perform DC analysis with C replaced by "open" ($I_C(\infty) = 0$) and L replaced by "short" ($V_L(\infty) = 0$). Find $V_C(\infty)$ and $I_L(\infty)$ and other required voltages and currents
3) $t = 0^+$ (switch in final position), Perform DC analysis with C replaced by voltage source having $V_C(0^+) = V_C(0^-)$ and with L replaced by current source having $I_L(0^+) = I_L(0^-)$. Find $I_C(0^+)$ and $V_L(0^+)$ and other required voltages and currents.
4) With switch in final position, calculate $\alpha$.

$\alpha = \frac{1}{2RC}$ 

$\alpha = \frac{R}{2L}$ 


5) Calculate $\omega_0 = \frac{1}{\sqrt{LC}}$

## (TRANSIENT RLC CONTINUED)

6) Select $f(t)$ equation type, where $f(t)$ is the required $v(t)$ or $i(t)$.

if $\alpha > w_0$: *Overdamped* $f(t) = A_1e^{s_1t} + A_2e^{s_2t} + f(\infty)$

$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$ $s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$

if $\alpha = w_0$: *Critically Damped* $f(t) = e^{-\alpha t}(A_1t + A_2) + f(\infty)$

if $\alpha < w_0$: *Underdamped*

$f(t) = e^{-\alpha t}(A_1 cosw_d t + A_2 sinw_d t) + f(\infty)$

$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

7) Form ${df(t)}/{dt}$ and substitute ${df(t)}/{dt}$ using

$v_L(t) = \mathrm{L}\frac{di_L(t)}{dt}$ $i_c(t) = \mathrm{C}\frac{dv_C(t)}{dt}$

8) Using initial conditions at t = $0^+$ and equations from (6) and (7), solve for all unknowns.

## DERIVATIVES FOR TRANSIENT RLC

if $f(t) = A_1e^{s_1t} + A_2e^{s_2t}$ then $\frac{df}{dt} = s_1A_1e^{s_1t} + s_2A_2e^{s_2t}$

if $f(t) = e^{-\alpha t}(A_1t + A_2)$ then $\frac{df}{dt} = -\alpha e^{-\alpha t}(A_1t + A_2) + e^{-\alpha t}(A_1)$

if $f(t) = e^{-\alpha t}(A_1 cosw_d t + A_2 sinw_d t)$ then

$\frac{df}{dt} = -\alpha e^{-\alpha t}(A_1 cosw_d t + A_2 sinw_d t)$
$+ e^{-\alpha t}(-w_d A_1 sinw_d t + w_d A_2 cow_d t)$

## TRIGONOMETRIC IDENTITES

$A cos(\alpha) + B sin(\alpha) = \sqrt{A^2 + B^2}\, cos\left(\alpha + tan^{-1}\frac{-B}{A}\right)$

$cos(\alpha \pm 90^0) = \mp sin(\alpha)$

$sin(\alpha \pm 90^0) = \pm cos(\alpha)$

$-cos(\alpha) = cos(\alpha \pm 180^0)$

## SINUSOIDAL EXCITATION

**Inductor:** current lags voltage (-$90^0$) or voltage leads current (+$90^0$), same thing.

**Capacitor:** current leads voltage (+$90^0$) or voltage lags current (-$90^0$), same thing.

**Resistor:** current and voltage are in-phase ($0^0$ phase difference).

## IMPEDANCE

Impedance of L: $\widehat{Z_L} = j\omega L$ Impedance of C: $\widehat{Z_C} = \frac{1}{j\omega C} = \frac{-j}{\omega C}$

Resonance Frequency, $\omega_0 = \frac{1}{\sqrt{LC}}$

Average Power, P, for sinusoidal waveforms ($cos\omega t$):

P = $\frac{1}{2}V_{peak}I_{peak}\cos(\theta_V - \emptyset_I)$ (time domain)

P = $V_{RMS}I_{RMS}\cos(\theta_V - \emptyset_I)$ (time domain)

P = $\frac{1}{2}Re\{\hat{V}\hat{I}^*\}$ (phasor)

$j = \sqrt{-1}$ $j^2 = -1$ $\frac{1}{j} = -j$

| Given | Convert to | using |
|---|---|---|
| $\hat{Z} = R + jX$ | $\hat{Z} = r\angle\emptyset$ | $r = \sqrt{R^2 + X^2}$ <br> $\emptyset = \tan^{-1}\frac{X}{R}\ (R \neq 0)$ |
| $\hat{Z} = r\angle\emptyset$ | $\hat{Z} = R + jX$ | $R = r\,cos(\emptyset)$ <br> $X = r\,sin(\emptyset)$ |

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